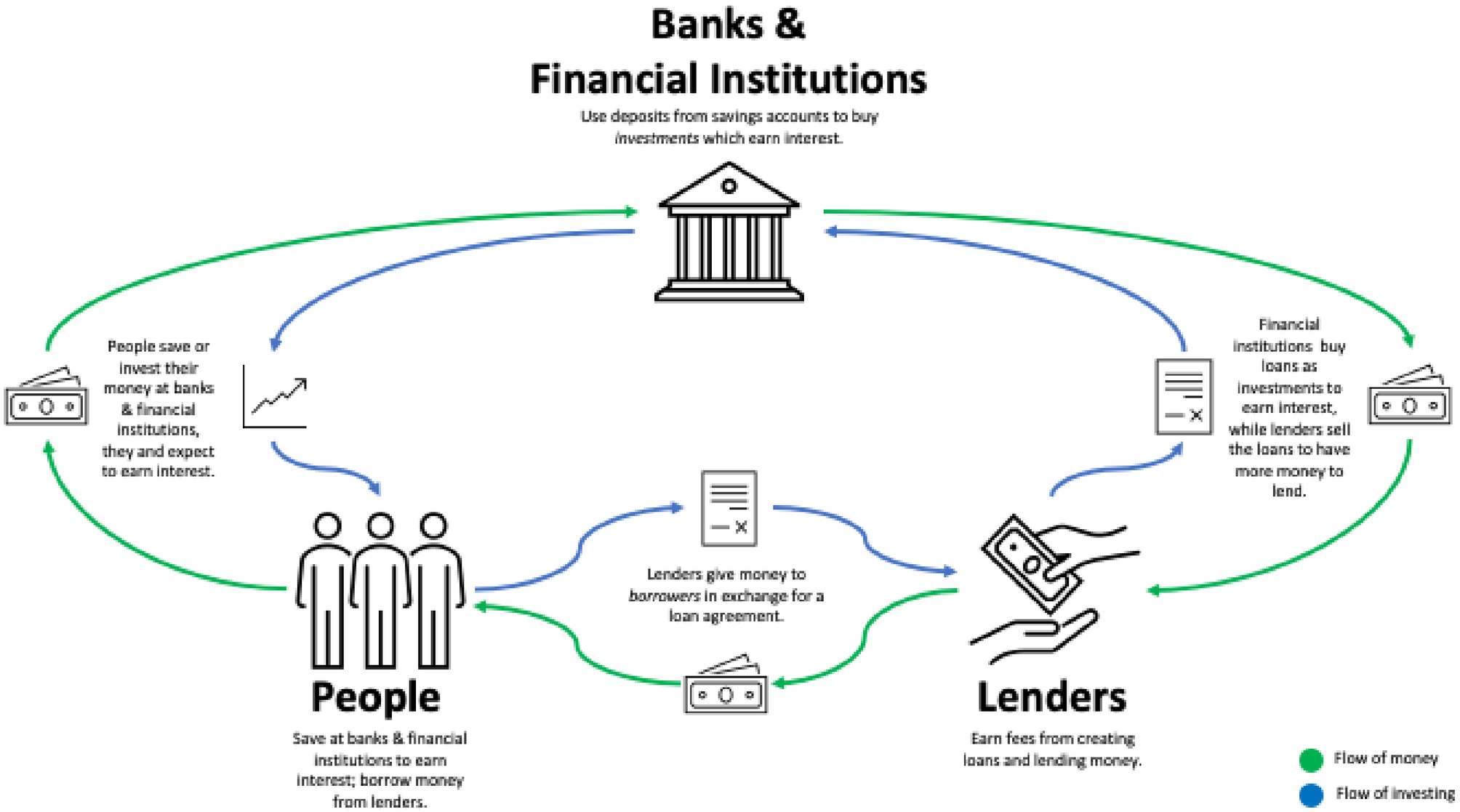
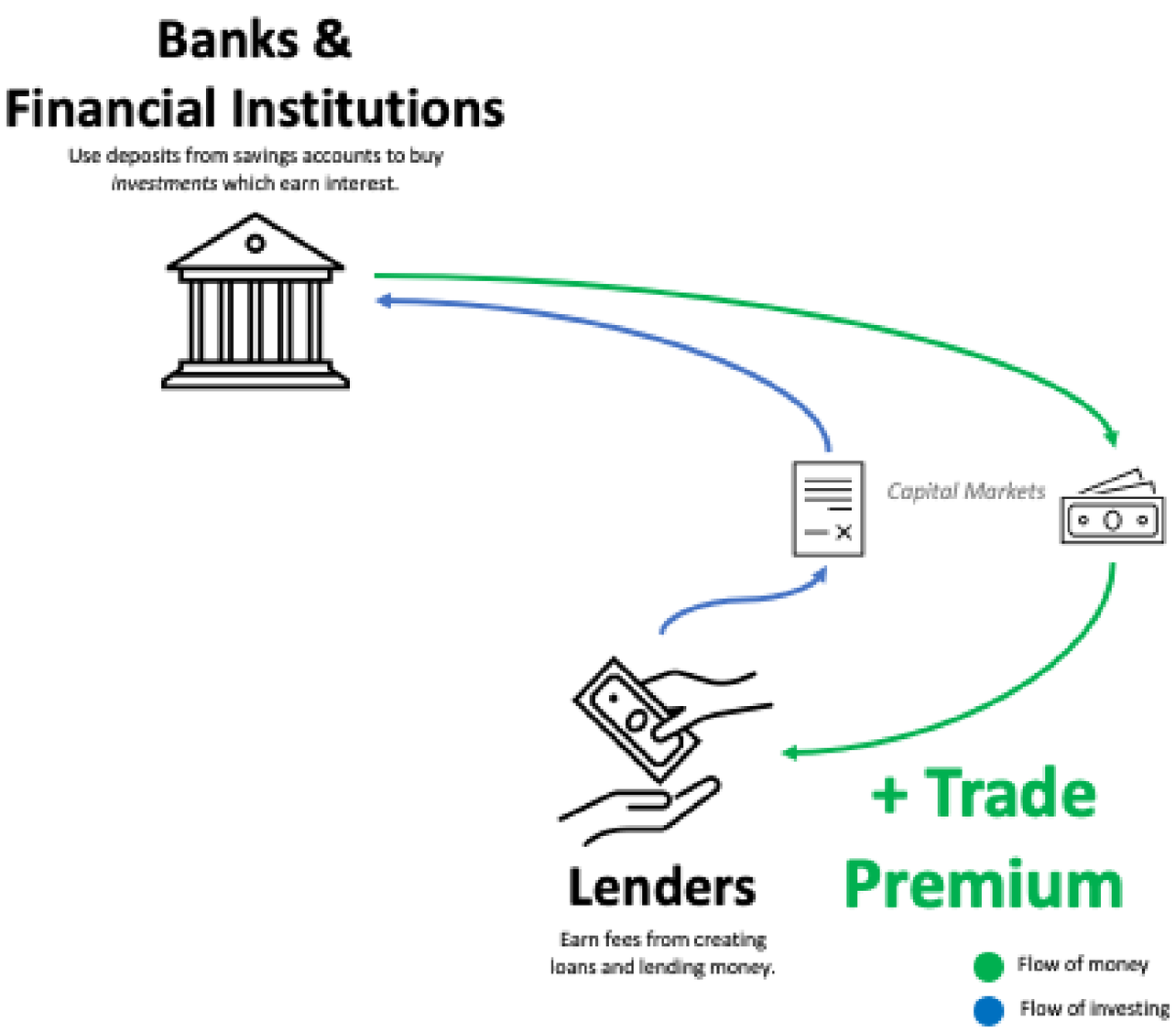
# Pricing in profit

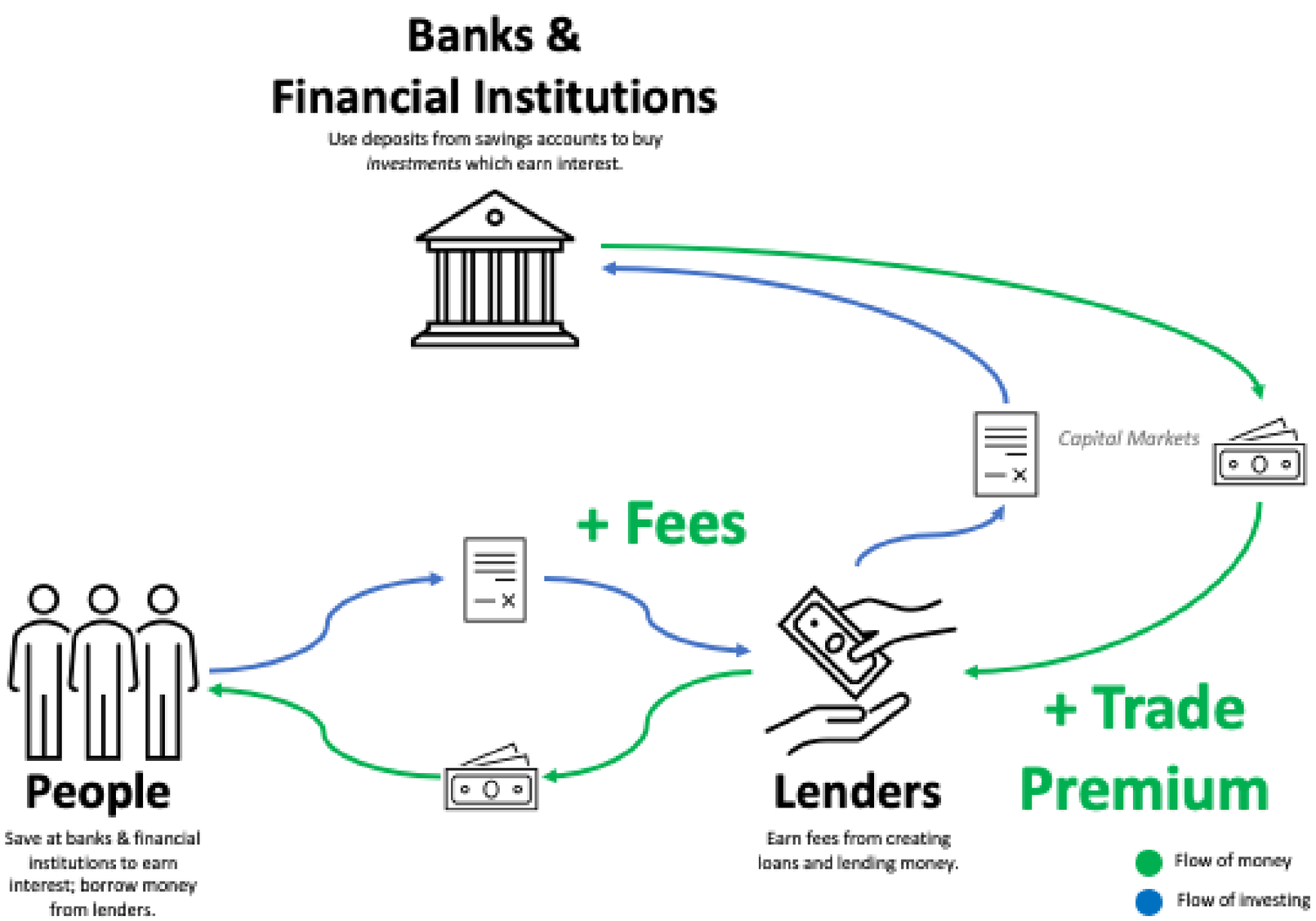
CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI

**Nick Edwards**
Capital Markets Analyst at Mynd

datacamp


Banks &
Financial Institutions
Use deposits from savings accounts to buy investments which earn interest.
People save or invest their money at banks & financial institutions, they and expect to earn interest.
Financial institutions buy loans as investments to earn interest, while lenders sell the loans to have more money to lend.
Lenders give money to borrowers in exchange for a loan agreement.
People
Save at banks & financial institutions to earn interest; borrow money from lenders.
Lenders
Earn fees from creating loans and lending money.
Flow of money
Flow of investing


Banks &
Financial Institutions
Use deposits from savings accounts to buy investments which earn interest.
Capital Markets
Lenders
Earn fees from creating loans and lending money.
+ Trade Premium
Flow of money
Flow of investing


Banks &
Financial Institutions
Use deposits from savings accounts to buy investments which earn interest.
Capital Markets
+ Fees
People
Save at banks & financial institutions to earn interest; borrow money from lenders.
Lenders
Earn fees from creating loans and lending money.
+ Trade Premium
Flow of money
Flow of investing

# Competitive markets

- A **competitive market** is a market where there are many buyers and sellers for an identical or very similar product
  - 4,338 lenders and 23.3mm loan applications in 2021 (U.S.)
  - No single buyer or seller can set a price

[1] https://www.consumerfinance.gov/data-research/hmda/summary-of-2021-data-on-mortgage-lending/

# Competitive markets: lenders

- Highly competitive markets are price sensitive!
- Lenders are constantly repricing to be competitive

# What is profit?

- **Profit** money made after all expenses have been paid.
  - `Revenue - Expenses`
- **Revenue** is the total money made from doing business
  - i.e., fees, trade premium
- **Expenses** are the costs of doing business
  - i.e., paying the mortgage banker, office expenses, discounts

**Example**: If revenue is 50,000 and expenses are 30,000 - what is the profit?

```
Profit = 50,000 - 30,000 = 20,000
```

# What is loan margin?

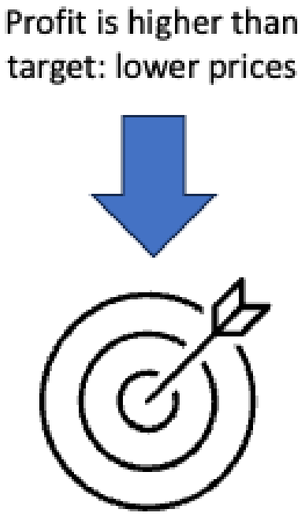
- **Loan profit margin** is the percent of the profit earned per dollar lent
  - `Profit ÷ Loan Amount`
- **Example:** What is the loan profit margin of a loan with 14,000 in profit and a loan amount of 100,000?
  - `14,000 ÷ 100,000 = 14%`

# Pricing and target profit

- **Business set margin targets**
  - Analyze costs and predict revenue

# Pricing and target profit

- **Business set margin targets**
  - Analyze costs and predict revenue
- **Higher than expected margins**
  - Not always good in a competitive market
  - Losing market share/customers to competition due to high pricing
  - Need to charge less



# Pricing and target profit

- **Business set margin targets**
  - Analyze costs and predict revenue
- **Higher than expected margins**
  - Not always good in a competitive market
  - Losing market share/customers to competition due to high pricing
  - Need to charge less
- **Lower than expected margins**
  - Risks going out of business!
  - Need to charge more
- **Financial markets impact lenders' profit!**

# Let's analyze!

CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI

# Congratulations!

## CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI

**Nick Edwards**
Capital Markets Analyst at Mynd

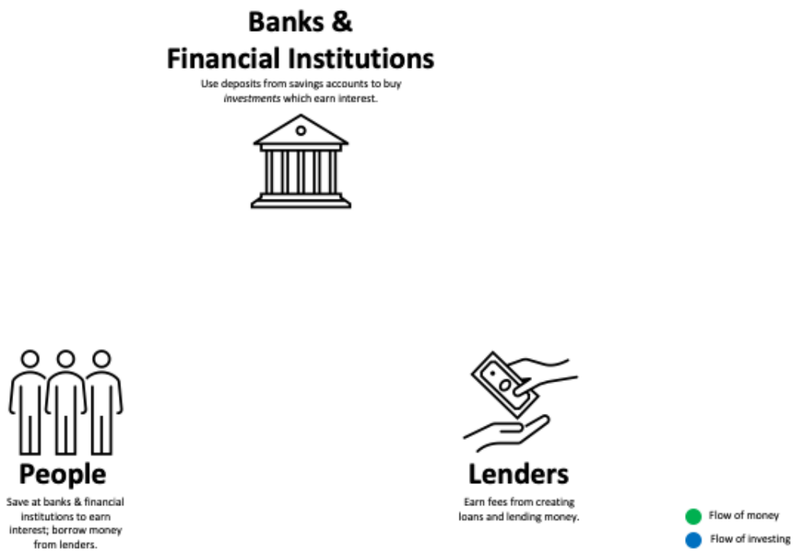
datacamp

# This was tough!

- Advanced case study
  - Complex concepts in the **financial system**
  - Advanced *Power BI* skills
- You became a trader in the **capital markets**

# In chapter 1 you...

- Learned how money moves from **people**, **banks**, and **lenders** in the financial system
- Learned the basics of **mortgage** and **loan agreements**
- Used *Power Query* to prepare the datasets
- Created complex `IF()` statements



# In chapter 2 you...

- Learned about trading mortgages in the **capital markets**
  - **Whole loan** and **Mortgage-backed securities**
- Created a **benchmark** based on **UMBS prices**
- Used *Power Query* and M to transform multiple datasets into a usable format for trade analysis
- Calculated **trade premiums** to determine the **total revenue** earned on the trade
- Calculated the **weighted average trade**

# In chapter 3 you...

- Learned about **competitive markets**
- Calculated **loan profit margin**
- Used AI-driven tools to find influencers in the data
- Determined what kind of pricing decisions should be made

# Best of luck!

CASE STUDY: MORTGAGE TRADING ANALYSIS IN POWER BI